

Mahammad Iqbal Shaik

Charlotte, NC (willing to relocate) | [LinkedIn](#) | 716-939-7031 | mahammad@buffalo.edu | [GitHub](#)

AI/ML Engineer with **4+ years of experience** designing and deploying scalable AI solutions in cloud-native environments. Specialized in Generative AI applications using cutting-edge Foundation Models, with a strong focus on solving real-world business problems through AI automation, productivity optimization, and inference efficiency. Skilled in deploying ML workflows on AWS and Azure using TensorFlow, PyTorch, MLflow and OpenCV. Proficient in RAG frameworks, MLOps pipelines and LLM deployments aligned with enterprise level standards.

WORK EXPERIENCE

Research Assistant | University at Buffalo, NY

February 2025 - Present

- Developed a specialized evaluation benchmark to uncover inherent limitations and "blind spots" in Large Language Models (LLMs), specifically assessing their personality traits and reasoning capabilities.
- Fine-tuned Llama-3 70B (4 bit QLoRA adapters) on 800 K persona rich dialogues, raising personality conditioned reasoning accuracy from 65 % to 72 % (+7 pp) while updating only 0.02 % of the model's weights.

ML/AI Engineer Intern | Order Appetit, NY

September 2024 - December 2024

- Drove operational efficiency by enabling non-technical users to query complex business data through an Agentic AI conversational system.
- Improved data accuracy to 95% by resolving 2,700+ product naming inconsistencies via a semantic search pipeline.
- Deployed a scalable backend for conversational AI in FastAPI, supporting 200+ users and integrating semantic similarity search with automated MongoDB and Pinecone data pipelines to boost accuracy and stakeholder efficiency.

Senior Software Engineer (R&D) | Next Education, India

April 2022 - July 2023

- Integrated a real time Grammar Error Correction service that profiles students' grammar gaps and feeds tailored exercises into the LMS, shrinking feedback loops from days to less than 1 hour for 2 K+ students.
- Built scalable, reproducible MLOps workflows with DVC for versioning, MLflow for experiment tracking, and Airflow for containerized data generation.
- Engineered an end-to-end **ETL pipeline** processing 150K+ schools with 20+ targeting parameters, achieving 90% reduction in lead qualification time through real time analytics and automated data quality checks.
- Boosted grading speed and accuracy by developing an SVM-based character recognition system for MCQ answer sheets, enabling rapid and reliable student assessment at scale.
- Reduced substitution errors by 40% on student answer sheets through a handwriting recognition system combining OpenCV, and NLP post-processing techniques. Mentored two interns in developing this solution.

Software Engineer | United Online, India

June 2020 - March 2022

- Enhanced system responsiveness and user experience by developing and maintaining RESTful APIs with Spring Boot for streamlined profile management and real-time updates.
- Delivered real-time profile and billing updates across web and mobile via Server-Sent Events (SSE) integration.
- Implemented Redis caching to cut DB load and boost API speed by 40%.
- Improved billing reliability by orchestrating complex workflows with a lightweight Saga pattern, handling multi-step processes like plan upgrades and partial-period charge adjustments.
- Ensured reliable deployments and code quality by building CI/CD pipelines with GitHub Actions, integrating Testcontainers and WireMock for automated testing.

SKILLS & INTERESTS

Languages: Python, Java, C++, Javascript, SQL

Databases: MySQL, MongoDB, PostgreSQL, Redis, DynamoDB

Frameworks: PyTorch, LangChain, LangGraph, Crew AI, HuggingFace, FastAPI, SpringBoot, React JS, AutoGen

ML/DL: XGBoost, Scikit-learn, Computer Vision, NLP, Predictive Modeling, Supervised and Unsupervised modeling

Gen AI: RAG, Pinecone, Milvus, FAISS, LLM Fine-tuning (PEFT, LoRA), Preference Alignment (DPO, GRPO)

MLOps / LLMOps: Git, Docker, Jenkins, Terraform, ZenML, MLFlow, vLLM, Azure AI Foundry, AWS Bedrock

PROJECTS

LLM Twin Project

- Fine-tuned Llama-3.1-8B via Direct Preference Optimization in a ZenML-powered RAG workflow, processing 500+ technical articles to boost domain-specific retrieval MRR by 15%.

LangQ: AI-Powered Interview Preparation Curator

- Developed an AI-powered Streamlit web application that transforms PDFs into interactive Q&A pairs using LangChain, OpenAI's GPT-4, FAISS, and PyPDF2, enhancing interview prep efficiency.
- Designed scalable components for embedding search and dynamic answer generation using LLMops, Python, and Streamlit, providing personalized learning experiences from unstructured documents.

EDUCATION

Masters in AI, SUNY University at Buffalo, NY (CGPA : 3.97)

August 2023 - January 2025

Bachelors in CS, University College of Engineering Kakinada

August 2016 - May 2020